



Hybrid Optimization Enabled Deep-Learning for Prostate Cancer Detection

Siva Kumar Reddy¹ · Kalaivani Kathirvelu¹

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Abstract

Prostate cancer (PCa) is now one of the most common types of cancer in gents and one of the most important causes of death in the United States. The best non-invasive imaging technique for diagnosing PCa is Magnetic Resonance Image. Each step in the conventional methods is optimized without allowing for error tolerance, resulting in significant computational costs. Hence, a Developed LHFGSO-DMN for PCa detection is introduced to reduce the proportion of PCa deaths. Modifying the objective function with dice coefficient and pixel-wise cross-entropy creates the optimized multi-objective semantic segmentation model known as multi-objective SegNet. This model is trained using the proposed Light Henry Firefly Gas Solubility Optimization (LHFGSO). The LHFGSO is the integration of a Henry Firefly Gas Solubility Optimization (HFGSO) and Light Spectrum Optimizer, and HFGSO is the formation of Henry Gas Solubility Optimization and Firefly Algorithm. Additionally, the LHFGSO method is used to train the Deep Maxout Network (DMN), which is used for cancer detection. The introduced multi-objective SegNet model with DMN for PCa growth recognition strategy reached superior execution measures with an accuracy of 94.63%, sensitivity of 93.46% and specificity of 95.72%.

Keywords SegNet · Light henry firefly gas solubility optimization · Light spectrum optimizer · Henry firefly gas solubility optimization · Firefly algorithm

✉ Siva Kumar Reddy
sivakumarreddybojja@gmail.com

¹ Department of Computer Science and Engineering, Vels Institute of Science Technology and Advanced Studies (VISTAS), Chennai, Tamil Nadu, India

1 Introduction

PCa is the second most frequent malignancy in men and the fifth most common cause of cancer-related mortality globally. The disease PCa is characterized by abnormal cell growth, and if not treated, these abnormal cells can proliferate out of control and develop tumors that have the potential to metastasize or spread the other parts of the human body. PCa has the potential to grow quickly and spread, but most men experience relatively slow growth [1]. It was the third leading cause of cancer-related death in men in the United States in 2017, and by 2022, it is predicted that PCa will cause 268,490 new cases and 34,500 deaths in the United States alone [2, 3]. Although radical prostatectomy is a general treatment for contained PCa, 20% to 40% of patients experience biochemical repetition surgery [4]. A radiologist who specializes in the genitourinary area must perform prostate biopsies, which are enveloping procedures that may result in bleeding or infection; consequently, it is not a preferred diagnosis for patients. Elevated serum levels of Prostate-Specific Antigen (PSA) have been utilized as a common biomarker for needle biopsy to aid in the early detection of PCa. PSA is a protein produced by both normal and malignant prostate cells. The PSA test measures the concentration of PSA in the blood. For this test, a blood sample is sent to a laboratory for analysis. However, the PSA screening puts a significant number of men at risk of being overdiagnosed and given additional treatment [5], necessitating repeated biopsy procedures for many non-prostate cancer patients. Distinguishing between potentially clinically significant PCa and indolent PCa is difficult [6], so various techniques and tools are developed for PCa detection.

PCa can be found utilizing X-ray, which is a technique for clinical imaging that is used in radiology to create pictures of the body's life systems and physiological cycles. Radio waves, strong magnetic fields, and gradients of magnetic fields are all used in MRI scanners to produce organ images. The PCa detection process now includes MRI, and the use of a computerized tool known as computer power-assisted diagnosing (CAD) has improved PCa classification and detection results and demonstrated a promising future for computer-based techniques for disease detection and characterization. MRI texture and morphology reveal the effects of diseases, and so classification and diagnosis of diseases are made easier with this MRI feature [1]. It has been demonstrated that PCa diagnosis based on multiparametric MRI (mp-MRI) and biparametric MRI (bp MRI) is effective and results in a higher biopsy yield rate than standard systematic biopsy [7]. MRI can be used to guide needles into the prostate during a prostate biopsy. If prostate cancer is found, an MRI may be done to determine the stage of cancer. MRI scans can also show whether the cancer has spread outside the prostate gland to the seminal vesicles or other nearby structures. Different MRI sequences allow for the collection of additional data that frequently serves as a complement to one another: T2-weighted (T2w), T1-weighted (T1w), and Proton Density (PDw) protocols can be used to obtain anatomical data [8–10]. The tumour's vascular structure was also revealed by Dynamic Contrast-Enhanced (DCE) MRI. Multi-focal prostate tumours are unfortunately common, making accurate MRI prognoses even more challenging [11].

In recent days, various techniques in medical images developed for detecting PCa. The distinct characteristics are drawn from the physiological, biological, and neuro-physiologic contexts. Biomarker tests and a Digital Rectal Exam (DRE) are common ways to find PCa in a patient [10]. There are three types of these tests: those based on blood, those based on urine, and hybrid tests. The development and training of algorithms are the foundation of Machine Learning (ML), a subfield of data science and, more specifically, artificial intelligence that enables computers to learn from data and make predictions without the need for specific programming [12, 13], including morphological, Scale Invariant Feature Transform (SIFT), entropy, and texture. In addition, morphology, elliptical shapes, geometry, complex dynamics, noise-resistant properties, and texture properties were used to extract hybrid features [1, 14]. In addition, medical image analysis and Natural Language Processing (NLP) benefit from Deep CNN (DCNN)'s superior performance. Despite this, only a small number of studies have focused on techniques without using CNN to identify prostate cancer [1]. To advance the Computer Aided Diagnosis (CAD) tool for detecting cancer, the medical imaging research community has turned its attention to methods that are enabled by deep learning and the promising results of CNN in computer vision [15, 16].

The primary objective of this study is to develop a method for detecting PCa by combining DMN and multi-objective SegNet.

The primary involvement of the paper is expanded as below:

- Developed LHFGSO-DMN for PCa prediction: By modifying the SegNet's objective function with dice coefficient and pixel-wise cross-entropy, an improved multi-objective SegNet is presented. LHFGSO model, introduced optimization model, is also used to train multi-objective SegNet. Additionally, the PCa is detected by DMN and it is trained using the developed LHFGSO approach.

The residual part of the paper is elucidated as follows: Section two deliberates the literature study of conventional approaches with their drawbacks and uses. Section 3 looks at the optimized multi-objective SegNet with DMN that was created. In Sect. 4, the proposed scheme's results and discussion are evaluated, and section five concludes the paper.

2 Literature Review

This section explains the merits and limitations of the conventional methods for detecting PCa. Wang et al. [12] developed the stacking-driven ensemble learning method for the prediction of PCa. An irregular backwoods classifier-based stacking technique was investigated for model mixes for the joining of base students like DT. Here, Diagnostic rules derived from the constructed ensemble learning model are accurate and easy to understand using this approach; however, neither a reduction in diagnosis costs nor savings in medical resources is achieved. Yoo, S., et al. [2] introduced the DCNN for PCa detection. Moreover, this approach required to

addition of other CNN architectures such as 3D CNNs (to feed the 3D Diffusion-Weighted Imaging (DWI) and Recurrent Neural Network (RNN)-45 (to account for the sequentialability of lesions in neighboring slices) for efficient PCa identification. This technique was successful in avoiding overfitting and produced exceptional results even with minimal input images. However, this strategy could not perform external validation using a different organization's database to improve the performance of the DCNN model. Yu, X., et al. [7] devised a Panoptic method for MRI-based PCa detection. The attention module was used in this technique to best combine global and local image features. Combining similar data from disparate sources was a better way to implement this method. However, this technique is not suitable for other cancers, such as breast cancer, and brain tumours. Cao, R., et al. [6] developed the Focal-Net for the early detection of PCa detection. The initial MRI image was taken and pre-processed to remove input image noises. After that, for a successful training process, data augmentation was carried out. This approach was effective in predicting fine-grained Gleason score groups also jointly detecting PCa, but it failed to add an optimization algorithm in clinical practice for finding lesions. De Vente, C., et al. [4] developed the Neural Network (NN) for PCa identification. This technique split the task into multiple networks, thereby improving the PCa prediction, but it failed to use histopathology-based delineations for PCa detection. Abbasi, A.A., et al. [1] devised the DCNN for the identification of PCa. Using a transfer learning approach, the CNN model (GoogleNet) achieved the highest performance, but it had high computational complexity. Mehta, P., et al. [17] AutoProstate, for MRI-based automatic PCa detection, was introduced. Here, if there were multiple focal lesions with the highest Gleason grade, the focal lesion with the highest Likert or PI-RADS v2 score was chosen. This method needed to lower the rate of AutoProstate False Positives before it could be validated. Lapa, P., et al. [11] developed the Conditional Random Fields as RF-RNN to enhance PCa detection. This technique had high generalization ability for even small-size datasets, but it failed to investigate the combination of CRFs into a CNN for high variability.

2.1 Challenges

The challenges encountered by current PCa detection methods are discussed below,

- Stacking driven ensemble learning method was devised in [1] for the prediction of PCa. The constructed ensemble learning model's diagnostic system was found to be reliable and interpretable, but the method failed to perform on the large dataset.
- In [14], DCNN was developed for PCa detection and each patient's MRI data consists of several thousand windows of pixels, and this method was able to classify PCa at the patient level with acceptable performance. However, this strategy could not perform external validation using a different organization's database to improve the performance of the DCNN model.

- A panoptic scheme [3] was introduced for the prediction of PCa; the advantage of this model was that it made it easier to combine information from different complementary sources. Despite this fact, this method was unable to incorporate distinct semantic branches into instance segmentation.
- For achieving higher accuracy, the Focal-Net technique was devised in [8] for PCa identification. However, this approach did not permit the precise registration of whole-mount slices with mp-MRI images.
- In [18], Elephant Herding Optimization-based Hyper-parameter Convolutional Deep Belief Network (CDBN-EHO) was introduced for diagnosing prostate cancer and Gleason grading in histological images. However, it requires external validation and a modest size of the sample.
- The common challenge of PCa detection was that extracting the hidden features for attaining higher performance is challenging. Some feature extraction technique was implemented but did not give good results leaving a need for further improvement.

The proposed LHFGSO-DMN approach obtained a high accuracy as it correctly determines the ratio of classified samples to total samples. Then, the proposed approach is successful in avoiding overfitting issues and produced better results even with minimal input images.

3 Proposed PCa Detection Using LHFGSO-DMN for PCa Prediction

This part elucidates an LHFGSO-DMN model for PCa detection. Primarily, the input MRI image is taken from a dataset [19], which is allowed for pre-processing using adaptive median filtering [20], and from the filtered image, ROI extraction [21] is carried out. Further, the segmentation process is done by Multi-objective SegNet [22], which divides an MRI image into areas based on cancer-affected regions. SegNet is trained using the Light Henry Firefly Gas Solubility Optimization (LHFGSO) algorithm, which is formed by the hybridization of HFGSO and LSO [23], where, HFGSO is produced by the combination of HGSO [24] and FA [25]. After that data augmentation, such as rotation, cropping, flipping, and random erasing is carried out, and is utilized for the identification of disease-affected areas. Then features, like LBP [26], SLBT [27] and statistical features [28] and, statistical features like, mean, variance, kurtosis, skewness, and entropy are determined, where feature extraction is used to reduce the data's dimensionality. Finally, cancer is detected using DMN [29], which is trained by LHFGSO. Figure 1 shows the developed PCa detection strategy in its schematic form.

3.1 Input MRI Image Acquisition

The input PCa image is acquired from a database [19], which is expressed as follows,

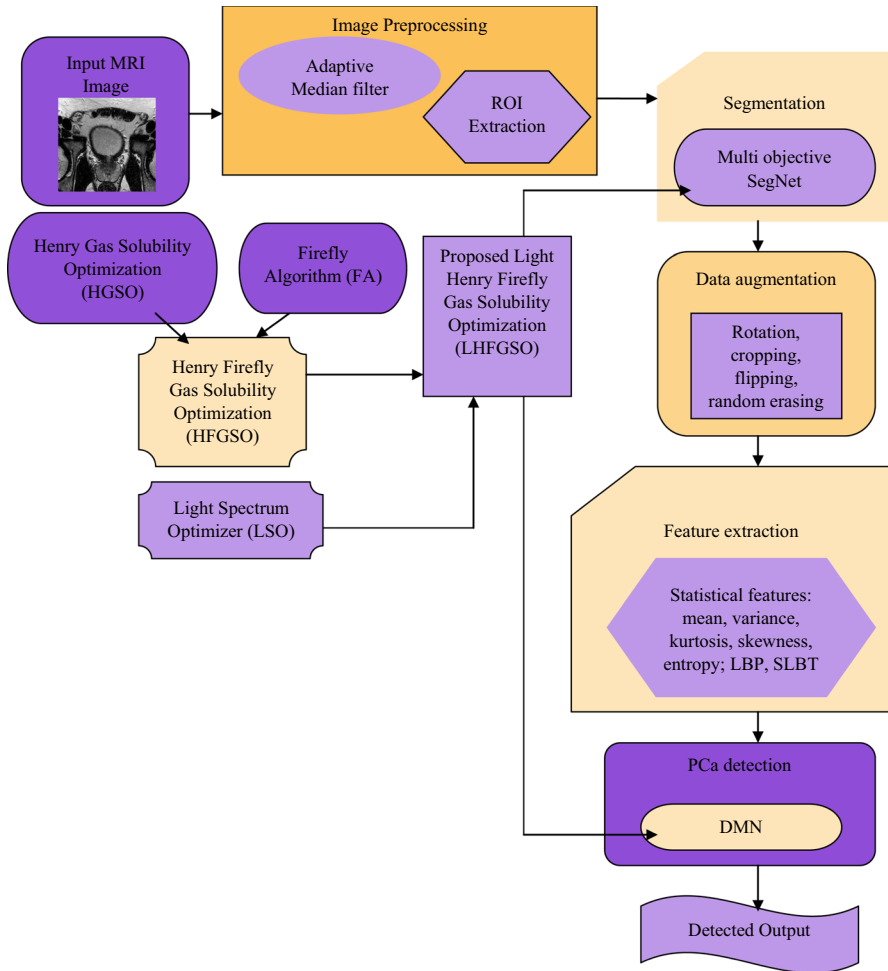


Fig. 1 Schematic view of hybrid optimization enabled deep learning for PCa detection

$$A = \{J_1, J_2, \dots, J_e, \dots, J_c\} \quad (1)$$

where the input image at e th index is J_e , the total number of the image is c , and a database is represented as A , J_e is applied to the input of preprocessing.

3.2 Preprocessing

The input image J_e is given to the preprocessing step, where ROI and an adaptive median filter are used for image preprocessing. During image processing, negative

and positive impulses brought on by errors or noisy channels frequently corrupt images, so error-free and noise-free image is received.

3.2.1 ROI Extraction

In ROI [21] reduces computational difficulty by extracting a specific area of the image. The pixel densities of input are taken into consideration when determining the ROI region. For effective segmentation and detection, the different portions and the dispersed segment are separated. The pixel force rate restricts the intrigued parts of the information picture, which is how the ROI is defined. External misrepresentations are eliminated by eliminating outer parts of the input.

3.2.2 Filtering

Here, an adaptive median filter [20] is used for the filtering process, and this filter is intended to preserve sharpness while removing mixed impulses with a high probability of occurrence. When it comes to preserving fine details while changing a high density of collaborative impulsive and non-impulsive noise, the adaptive median filter is simpler and works better, and the output of the pre-processed image obtained has indicated as B_e .

3.3 Segmentation

After finishing pre-processing, segmentation is carried out utilizing the enhanced multi-objective SegNet model [22], which is trained by LHFGSO. The process of dividing an image into several segments is known as segmentation. Segmented output is utilized to detect cancer effectively, so the pre-processed image B_e serves as the SegNet model's input, where the output of segmentation is C_t .

3.3.1 Multi Objective SegNet

By combining SegNet structure with objective functions like the dice coefficient and cross-entropy, where optimized multi-objective SegNet [22] method is presented.

(i) SegNet Architecture

The SegNet model [22] is an encoder structure with a decoder network that follows the final pixel-based classification layer. Typically, the architecture of the encoder consists of thirteen Conv layers where weights are used for training and has 13 layers of decoder network. An encoder consists of convolution with an element-wise tanh non-linearity, filter bank, sub-sampling and max-pooling to obtain the feature maps. Additionally, feature maps with high resolution at deeper encoder results are created using fully connected layers. Thus, the SegNet encoder

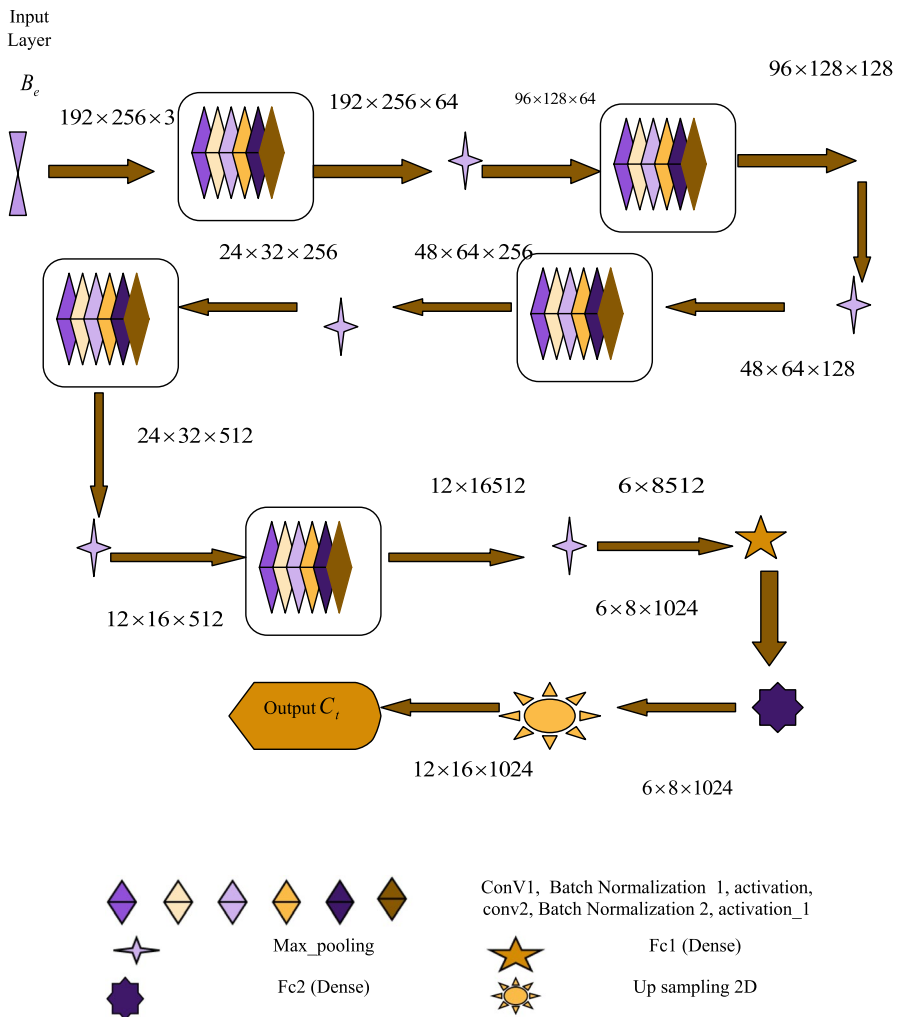


Fig. 2 Architecture of SegNet

structure's number of parameters decreases at deeper encoder output and larger decoders increase the performance of a given encoder network. The SegNet is the more efficient model as it uses maximum pooling indices to upsample and convolves a trained decoder filter bank without feature-based learning maps. Figure 2 depicts the architecture of SegNet.

3.3.2 Loss Function

As a pixel-level classification task, semantic segmentation frequently employs Mean Square Error (MSE) or cross-entropy loss as its loss function [30]. Coming up next is a clarification of these two capabilities,

a) Pixel-Wise Cross-Entropy

The loss calculates the average result for each pixel without taking into account the class imbalance problem, which is expressed as,

$$D(a, b) = \sum_d a_d \log b_d \quad (2)$$

where the predicted label is a_d , d indicates the category and the true label is b_d .

b) Dice Coefficient

In addition, the dice coefficient represents overlapping between two samples and is used to estimate the performance of segmentation. The formula for the dice coefficient is,

$$B(a, b) = \frac{2 \sum_d b_d \times a_d}{\sum_d b_d + \sum_d a_d} \quad (3)$$

In the end, the dice coefficient and the pixel-wise cross-entropy are combined in the following way:

$$E(a, b) = (1 - \sigma) \left[\sum_d a_d \log(b_d) \right] - \sigma \log \left(\frac{2 \sum_d b_d \times a_d + F}{\sum_d b_d + \sum_d a_d + F} \right) \quad (4)$$

where smooth is F , which, for example, is a small number $1e - 15$ and $\sigma = 0.75$.

3.3.3 LHFGSO for Training SegNet

The LHFGSO model, an optimization technique, is used to train the created multi-objective SegNet model. The proposed LHFGSO is an incorporation of HFGSO and LSO [23], where the HFGSO is the integration of HGSO [30] and FA [25]. There are primarily two primary functions in the FA: attracting a potential grey and a partner for mating. Blazing fills in as a defensive advance notice model for this situation. In addition, the fact that all fireflies are male or female is considered, and their magnetism is inversely relative to their intensity. When it came to resolving a multi-objective optimization function, this model was ineffective. So, HGSO is utilized for integration issues. Based on Henry's law, the metaheuristics technique known as the HGSO model was developed. The quantity of a particular gas that melts a particular kind of liquid at the current temperature is determined by Henry's law, which is an essential law. The rainbow effect, which occurs when sunlight passes through a drop of water, serves as the foundation for the LSO metaheuristic algorithm. LSO appears to have struck a good

balance between exploitation and exploration as a result. Following is an illustration of the algorithmic procedure of the designed LHFGSO technique:

(i) Initiate the Process

At first, the total gases and their locations are initialized during solution initialization, which is specified as

$$G_i(f+1) = G_{\min} + g \times (G_{\max} - G_{\min}) \quad (5)$$

where, random number with 0 and 1 is g , i^{th} gas in population location is G_i , and bounds of problem are $G_{\min}$ and $G_{\max}$, and f symbolizes iteration. The total amount of gas is i , $c(H_i(f))$ is a constant kind of value of Henry, $I_{i,c}$ which denotes the partial pressure of gas i in the cluster, $\Delta_{sol}K/L$ is a constant value of $c(M_i)$.

$$H_i(f) = h_1 \times rand(0, 1), I_{i,c} = h_2 \times rand(0, 1), M_d = h_3 \times rand(0, 1) \quad (6)$$

then, h_1, h_2, h_3 are constant parameters $h_1 = 5H - 02$, $h_2 = 100$, and $h_3 = 1H - 02$.

(ii) Clustering

Henry's constant values of the population agents are the same because they are divided into clusters that are the same size and have the same kinds of gases and because all of the clusters have the same gases.

(iii) Calculate the Fitness Function

By estimating the optimum fitness solution, which is calculated using Eq. (4), the fitness solution is used to determine the best solution.

(iv) Henry's Coefficient Updating

After estimating the fitness function, the following equation is utilized to update Henry's coefficient:

$$H_i(f+1) = H_i(f) \times \exp(-M_d \times (1/N(f) - 1/N^\psi)) ; N(f) = \exp(-f/j) \quad (7)$$

where the temperature is N , H_i is Henry's coefficient, the whole number of iterations is j , and N^ψ denotes constant, $N^\psi = 298.15$, $M_d = h_3 \times rand(0, 1)$.

(v) Solubility Updating

The solubility is updated following the completion of Henry's coefficient, and its expression is as,

$$O_{i,c}(f) = P \times H_i(f+1) \times I_{i,c}(f) \quad (8)$$

where $O_{i,c}$ signifies gaseous solubility of the gas i in the cluster c , P is a constant value, and the partial pressure on gas i in the cluster c represents as $I_{i,c}$.

(vi) Position Updating

The devised LHFGSO algorithm is used to update the location, as shown below:

$$q_{ok}(e+1) = \frac{(\omega_0 e^{-\tau_{ok}^2} Q_k(e) + \theta \beta_o)(Cs\tau + Cs\theta - 1) + (Cs(\tau Q_{o,best}(e) Q_{best}(e))(1 - \omega_0 e^{-\tau_{ok}^2})}{[Cs\tau + Cs\theta - \omega_0 e^{-\tau_{ok}^2}]} \quad (9)$$

Substituting $q_{ok}(e)$ on both sides,

$$q_{ok}(e+1) - q_{ok}(e) = \frac{(\omega_0 e^{-\tau s_{ok}^2} Q_k(e) + \theta \beta_o)(Cs\tau + Cs\theta - 1) + (Cs(\tau Q_{o,best}(e) Q_{best}(e))(1 - \omega_0 e^{-\tau s_{ok}^2}))}{[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2}]} - q_{ok}(e) \quad (10)$$

From LSO [25], the meteorological phenomenon of the colourful rainbow spectrum, which is caused by light dispersions passing through raindrops at various angles, serves as the inspiration for LSO, which solves continuous optimization issues.

$$\vec{q}_{e+1} = (\vec{q}_{s_1}^f + |GT_5| \times (xq_{s_2} - xq_{s_3})) \times \vec{t} + (1 - \vec{t}) * q_e \quad (11)$$

$$\vec{q}_{e+1} = q_{ok}(e+1) \quad (12)$$

$$\vec{q}_e = q_{ok}(e) \quad (13)$$

$$\vec{q}_{s_1}^f = q_{ok}^f s_1(e) \quad (14)$$

$$\vec{q}_{s_2} = q_{ok} s_2(e) \quad (15)$$

$$\vec{q}_{s_3} = q_{ok} s_3(e) \quad (16)$$

$$q_{ok}(e) = \frac{q_{ok}(e+1) - (q_{ok}^f s_1(e) + |GT_5| * (q_{ok} s_2(e) - q_{ok} s_3(e))) * \vec{t}}{(\omega_0 e^{-\tau s_{ok}^2} Q_k(e) + \theta \beta_o)(Cs\tau + Cs\theta - 1) + (Cs(\tau Q_{o,best}(e) Q_{best}(e))(1 - \omega_0 e^{-\tau s_{ok}^2}))} \quad (17)$$

Substituting Eq. (17) in Eq. (10)

$$q_{ok}(e+1) = \frac{(\omega_0 e^{-\tau s_{ok}^2} Q_k(e) + \theta \beta_o)(Cs\tau + Cs\theta - 1) + (Cs(\tau Q_{o,best}(e) Q_{best}(e))(1 - \omega_0 e^{-\tau s_{ok}^2}))}{[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2}]} - \frac{[q_{ok}(e+1) - (q_{ok}^f s_1(e) + |GT_5| \times (q_{ok} s_2(e) - q_{ok} s_3(e))) * \vec{t}]}{(1 - \vec{t})} + q_{ok}(e)$$

$$q_{ok}(e+1) + \frac{q_{ok}(e+1)}{1 - \vec{t}} = \frac{(1 - \vec{t}) + \left[(q_{ok}^f s_1(e) + |GT_5| \times (q_{ok} s_2(e) - q_{ok} s_3(e))) * \vec{t} \right] \left[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2} \right] + q_{ok}(e)(Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2})(1 - \vec{t})}{[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2}](1 - \vec{t})} \quad (19)$$

$$\begin{aligned}
& q_{ok}(e+1)[1-\bar{t}+1] = \\
& (1-\bar{t}) + \\
& \frac{\left[(q_{ok}^f s_1(e) + |GT_5| \times (q_{ok} s_2(e) - q_{ok} s_3(e)) * \bar{t}) \left[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2} \right] + q_{ok}(e)(Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2})(1-\bar{t}) \right]}{\left[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2} \right]}
\end{aligned} \quad (20)$$

$$\begin{aligned}
& q_{ok}(e+1) = \\
& (1-\bar{t}) + \\
& \frac{\left[(q_{ok}^f s_1(e) + |GT_5| \times (q_{ok} s_2(e) - q_{ok} s_3(e)) * \bar{t}) \left[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2} \right] + q_{ok}(e)(Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2})(1-\bar{t}) \right]}{\left[Cs\tau + Cs\theta - \omega_0 e^{-\tau s_{ok}^2} \right] [2-\bar{t}]}
\end{aligned} \quad (21)$$

Here, τ is scaling factor, s denotes the random constant, $Q_{best}(e)$ represents e^{th} iteration best solution, C is the flag that varying direction, $Q_{o,best}(e)$ the best solution in a cluster o , ω_0 is attractiveness at $s = 0$, β_o denotes a random integer vector derived from the Gaussian distribution, and $U_{ok}(e)$ signifies solubility of a gas o in the cluster k , θ is constant, $\bar{t}$ represents vector together with random values of $(0, 1)$, $q_{ok} s_1(e)$, $q_{ok} s_2(e)$ signifies solution taken randomly from the present population. GT_5 denotes the scalar of a number that is normally distributed.

(vii) Escape from Local Optimum

The following expression is used to rank and select the worst search agents because this is done to avoid the local optimal.

$$Q_k = Q \times (rand(l_2 - l_1) + l_1) \quad ; l_1 = 0.1 \text{ and } l_2 = 0.2 \quad (22)$$

where search agents' quantity is Q .

(viii) Search Agent's Position Updating

Search agent's locations are updated using the subsequent equation:

$$R_{i,c} = R_{\min i,c} + v(R_{\max i,c} - R_{\min i,c}) \quad (23)$$

where bounds of the problem are $R_{\max}$, and $R_{\min}$, $R_{i,c}$ indicates the location's gas i in the cluster c , and the random integer is v .

(ix) Feasibility of Solution Calculation

To find the best solution for each image, Eq. (4) is used to estimate the fitness measure. The most effective fitness solution is the one with the lowest value.

(x) Termination

The aforementioned procedures are accepted continuously awaiting the best resolution is found, and the procedure is completed after the greatest number of iterations has been obtained. Algorithm 1 specifies the devised HFGSO technique's pseudo-code.

Algorithm 1 LHFGSO's pseudo code

Pseudocode of developed LHFGSO model
Input: Total number of gas
Output: $G_i(f+1)$ is Optima solution
Begin
Initiate gas quantity and location.
Split the population agents into a few groups.
Recognize in cluster
Evaluate fitness solution
while $f < \max . number of iteration$ do
For each search agent do
Change the location of every search agent using an equation (21)
end for
Updation Henry's gas type coefficient using an expression (7)
Updating the gas's solubility using the equation. (8)
Using an equation (22), rank and select the worst agents
Update the worst agent's location using an equation (23)
end while
$f = f + 1$
Return $G_i(f+1)$
End

Moreover, For segmenting a pre-processed image for the subsequent process, the proposed LHFGSO achieved improved performance. Segmented output is indicated C_t , which is transferred to augmentation.

3.4 Data Augmentation

Segmented output has C_t which has been transferred to the augmentation process, where editing, flipping, pivoting and irregular deleting are handled. To improve detection performance, Data augmentation is a method for increasing the size of existing data without adding any new data.

3.4.1 Rotation

The image is rotated by -50 degrees to 30 degrees. when this model is applied to a segmented image and the outcome is denoted as F_1 .

3.4.2 Cropping

This method is used to select a specific region that reduces the segmented image by 30% on one side, which gives F_2 .

3.4.3 Flipping

An image can be rotated around its vertical, horizontal, or both axes when it is flipped [31]. Without requiring any counterfeit handling, it empowers clients to expand the all-out number of pictures in a dataset to its greatest. Here, the output of flipping is F_3 .

3.4.4 Random Erasing

This technique is not a part of the transformation of geometry [31]. Random erasing works by randomly erasing one square from the square region of the image and producing, which has obtained is denoted as F_4 .

Consequently, the overall output of the data augmentation is

$$F_O = \{F_1, F_2, F_3, F_4\} \quad (24)$$

Then, data augmented output F_o is transferred to the next phase.

3.5 Feature Extraction

To generate additional highly significant features, the feature extraction process performs a limited number of transformations on the actual features. The dimension of the feature space can frequently be decreased in feature extraction without sacrificing a significant amount of the information contained within the feature space itself. In this case, a segmented image F_O serves as the input for feature extraction. Structural features like LBP [26], and SLBT [27], and Statistical features [28] like mean, variance, kurtosis, skewness, and entropy are a few of the features that are possible to extract.

a) LBP: The global texture-enabled feature descriptor, LBP [26], allows these features to label a pixel's value by thresholding the area around each pixel and interpreting the results as binary numbers. The pixel value of LBP (β_m, θ_m), expressed as,

$$p_1 = LBP_{(\beta_m, \theta_m)} = \sum_{e=0}^7 D(\chi_e - \chi_m) * 2^e \quad (25)$$

Here, χ_e is an intensity of frame, $LBP(\beta_m, \theta_m)$, e denotes the 8 neighbours of the central pixel, and χ_e is used for aiming neighbour pixels. β_m as distribution of sample points on

a radius for θ_m and evaluates LBP, $D(q)$ is defined as, $D(q) = \begin{cases} 1 & \text{if } q \geq 0 \\ 0 & \text{if } q < 0 \end{cases}$, where LBP is denoted as p_1 .

b) SLBT:

In texture modelling, SLBT [27] projects the shape-free patch's LBP feature histogram into eigenface space. As a result, SLBT can extract shape modelling's global shape variation and texture modelling's local shape and texture variation more effectively, here SLBT is denoted as p_2 .

The structural features are denoted as

$$T_s = \{p_1, p_2\} \quad (26)$$

where, p_1 is LBP output and p_2 is SLBT output.

c) Statistical Features:

Here, the structural feature T_s is used to determine statistical features, where statistical features like, mean, variance, Kurtosis, skewness, and entropy are performed in the feature extraction [28].

(i) Mean: It reveals the concentration of a distribution's data. The region of interest's global temperature can be displayed using the mean.

$$\sigma = \sum_{o=0}^{S-1} o * D(o) \quad (27)$$

where, $D(o)$ indicates the o^{th} happening probability, the total amount of grey level is S , grey level of image $p(q, r)$ is represented as o , where the mean is denoted as σ_1 .

ii) Variance

The grey level deviation value concerning the mean grey level in the image is shown here, which is computed by,

$$\mu^2 = \sum_{o=0}^{S-1} (o - \sigma)^2 * D(o) \quad (28)$$

σ indicates the mean value, here σ_2 is the variance's output.

(iii) Kurtosis: Another name for kurtosis is "fourth normalized moment." It estimates a circulation's evening out corresponding to a typical dissemination, which is expressed as,

$$\tau_1 = \sigma^{-4} \left[\sum_{o=0}^{S-1} (o - \sigma)^4 * D(o) \right] \quad (29)$$

Here, kurtosis is denoted as σ_3 .

(iv) Skewness: A specific feature's degree of asymmetry around the mean is the subject of this term, and it is expressed as,

$$\tau_2 = \sigma^{-3} \left[\sum_{o=0}^{S-1} (o - \sigma)^3 * D(o) \right] \quad (30)$$

where skewness is indicated as o_4 .

(v) **Entropy**: It measures the system's disorder in addition to the information in the segmented images. The more symmetric the temperature distribution, the lower the entropy. People whose breast images (left and right) have significantly different entropy are more likely to have abnormalities and more asymmetry.

$$B = \sum_{o=0}^{S-1} D(o) * \log_2[D(o)] \quad (31)$$

Here, entropy is designated as o_5 .

Statistical feature output is denoted as, $I''_s = \{o_1, o_2, o_3, o_4, o_5\}$, and extracted features are denoted as $\hat{F}$.

3.6 PCa Detection Using Proposed LHFGSO-DMN

Feature-extracted output $\hat{F}$ is utilized for the PCa prediction process with hybrid optimization-based DMN. Here, DMN [29] is trained by the proposed hybrid optimization, namely LHFGSO. The proposed LHFGSO is a formation of HFGSO and LSO [23], where the HFGSO is the incorporation of HGSO [30] and FA [25].

3.6.1 DMN Architecture

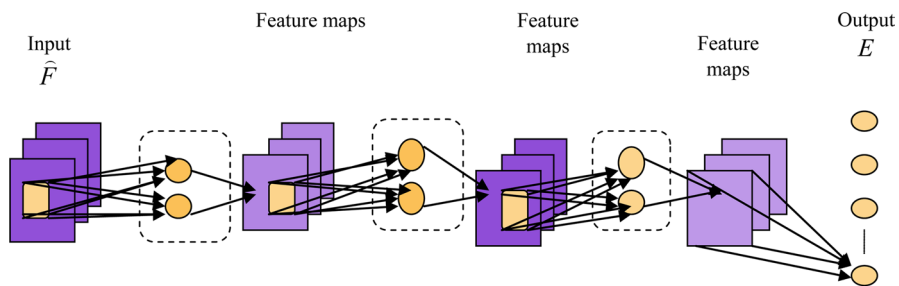
The multi-layer structured max out offers the benefits of a non-saturated activation function and a trainable activation function approximator. By partially preventing concealed units from entering dormant states, max-out facilitates optimisation. The max-out unit is not a random function approximator; rather, it is a trainable activation function. Multi-layer max-out networks are a class of trainable activation functions having a multilayer structure (MMN) The following formula determines whether a concealed unit is activated:

$$p_{o,k}^1 = \max_{k \in [1, a_1]} \hat{F}^I N \dots ok \dots + m_{ok} \quad (32)$$

$$p_{o,k}^2 = \max_{k \in [1, a_2]} p_{o,k}^1 {}^I N \dots ok \dots + m_{ok} \quad (33)$$

$$p_{io,k}^b = \max_{k \in [1, a]} f_{o,k}^{n-1} {}^I N \dots ok \dots + m_{ok} \quad (34)$$

$$p_{ok}^h = \max_{k \in [1, a_n]} f_{o,k}^{h-1} {}^I N \dots ok \dots + m_{ok} \quad (35)$$

**Fig. 3** DMN architecture**Table 1** Hyperparameters

Parameters	DCNN	Panoptic model	Focal-Net	ResNet	HFGSO based DRN	Proposed LHFGSO-DMN
Epochs	20	20	20	20	25	30
Batch size	64	32	64	32	32	32
Learning rate	0.001	0.001	0.001	0.001	0.001	0.001
Optimizer	SGD	Adam	Adam	RMSprop	HFGSO	LHFGSO

where DMN's total number of layers is h , m is the bias of the network, and N is the weight of the network, a_n is the total amount of units in the n^{th} layer. Conventional nonlinear activation functions can be accurately approximated by a DMN. when a is not lower than 2, like, the absolute value rectifier and rectified linear rectifier. By increasing the parameter a , Even though a more complex nonlinear activation function is required for potential feature extraction, a DMN could approximate any activation function. In the meantime, a secondary effect is that the calculation intricacy increments decisively, here Fig. 3 signifies the architecture of DMN.

3.6.2 Proposed LHFGSO for Training DMN

The developed LHFGSO model is used to train DMN. The proposed LHFGSO is a formation of HFGSO and LSO [23], where the HFGSO is the integration of HGSO [30] and FA [25]. Section 3.3.3 explains the LHFGSO creations, and the fitness solution of the DMN is expressed as,

$$MSE = \frac{1}{h} \sum_{o=1}^h (I_o^* - I_o)^2 \quad (36)$$

where h is the number of samples, I_o^* is the expected output, and DMN output is I_o .

4 Results and Discussion

The proposed LHFGSO-DMN PCa detection model is discussed in this section, which presents the findings. This section also describes the experimental arrangement, evaluation metrics, explanation of the dataset, comparative methods, analysis, and discussion.

4.1 Experimental Set-Up

The devised PCa detection approach is implemented in the PYTHON tool using the Prostate MRI dataset [19]. Table 1 shows the hyperparameters used by all the methods.

4.2 Dataset Description

The Prostate MRI dataset is used to carry out the designed PCa detection method [19]. The prostate MRI data come from the records of patients. The database contains the examination type, explanation, and date of 230 patients' records, as well as information about the disease and its detection, where twenty patients' medical records were used in the new approach. The Prostate MRI dataset consists of a total parameter of 230,931 (902.07 KB), out of which 230,611(900.82 KB) parameters are trainable and 320 (1.25 KB) are non-trainable parameters.

4.3 Evaluation Measures

Three performance evaluation metrics, such as accuracy, specificity, and sensitivity are used to estimate the performance of the proposed LHFGSO-DMN.

(i) Accuracy

The accuracy is used to determine the ratio of correctly classified samples to total samples and is expressed as,

$$\Omega = \frac{I_q + I_k}{I_q + I_k + Q_q + Q_k} \quad (37)$$

(ii) Specificity

The specificity, which is denoted by, is intended for predicting precise detection rate of true positives,

$$\wp = \frac{I_q}{I_q + Q_k} \quad (38)$$

(iii) Sensitivity

In PCa detection, sensitivity is calculated to predict the precise rate of true positives and is indicated by,

$$\lambda = \frac{I_k}{I_k + Q_q} \quad (39)$$

Here, I_q is the True Positive, I_k denotes the True negative, Q_q specifies the false positive rate, Q_k implies the false negative, Ω denotes the accuracy, ϕ indicates specificity, and refers to sensitivity.

4.4 Comparative Techniques

The different comparative techniques used to analyse the effectiveness of the proposed PCa detection method are DCNN [2], Panoptic model [7], Focal-Net [6], ResNet [32], HFGSO-based Deep Residual Network (DRN) for PCa prediction.

DCNN: This approach required to addition of other CNN architectures for efficient PCa identification. This technique was successful in avoiding overfitting and produced exceptional results even with minimal input images.

Panoptic Model: It is a panoptic method for MRI-based PCa detection. The attention module was used in this technique to combine global and local image features. Combining similar data from disparate sources was a better way to implement this method.

Focal-Net: It is the Focal-Net model for the early detection of PCa detection. This approach was effective in predicting fine-grained Gleason score groups also jointly detecting PCa.

ResNet: A residual network was implemented based on segmentations from an expert radiologist on T2-weighted, apparent diffusion coefficient map, and high b-value diffusion-weighted images. Mp-MRIs from 346 patients were used in this study.

Henry Firefly Gas Solubility Optimization based Deep Residual Network (HFGSO based DRN): HFGSO-based DRN for Prostate cancer detection is implemented here. In the proposed method, the weights and bias of DRN are optimized using HFGSO for better simplification and classification of large-value gene datasets.

4.5 Experimental Output

This section provides an explication of the introduced PCa identification approach's experimental results. Figure 4a portrays the sample of input MRI images 1 and 2, ROI image-1 and 2 is depicted in Fig. 4b Pre-processed images 1, 2 are elucidated in Fig. 4c, Consequently Fig. 4d–g, depicts the flipping, cropping, rotation, Random erasing output image of input MRI image-1 and 2, Fig. 4h. indicates the features extracted output of MRI images 1 and 2.

Fig. 4 Experimental result of proposed PCa detection using proposed LHFGSO-DMN **a.** Input image, **b.** ROI extracted image, **c.** pre-processed image, **d.** flipping image, **e.** cropping image, **f.** rotation image, **g.** random erasing image, **h.** feature extracted output

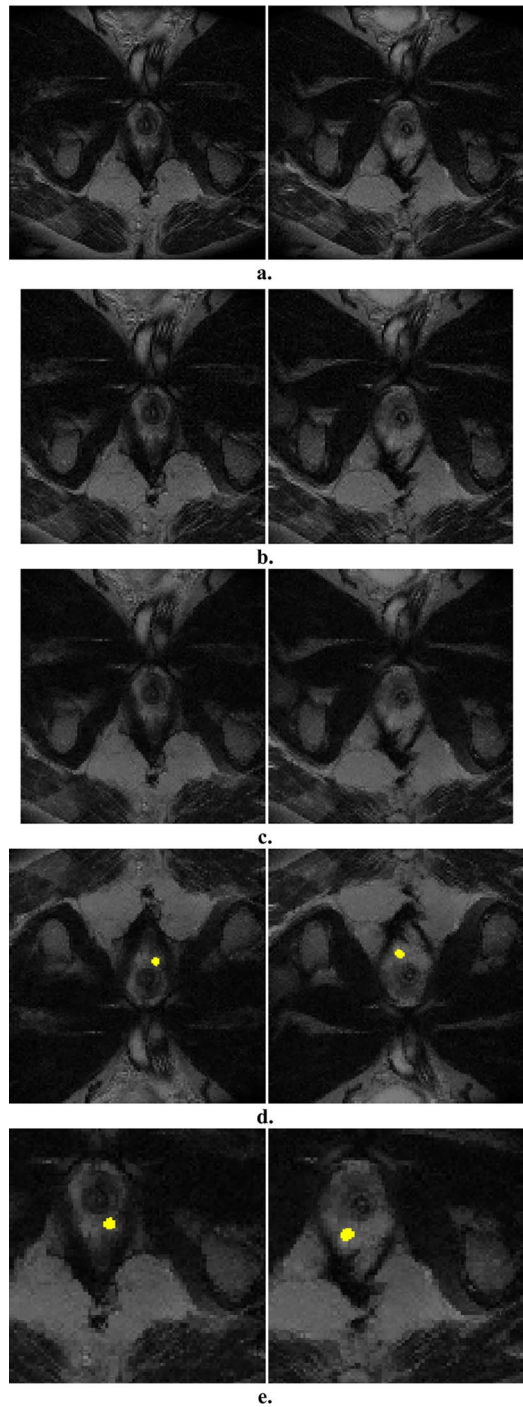
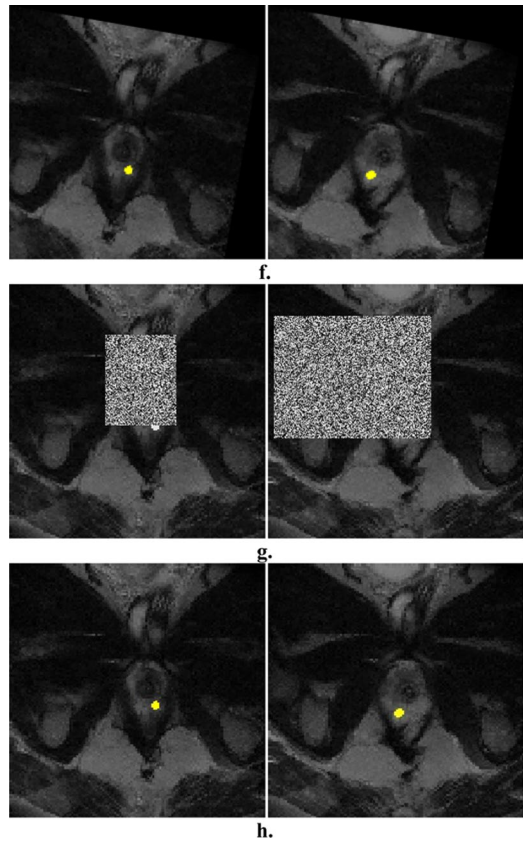


Fig. 4 (continued)



4.6 Comparative Assessment

Varying training data and k-fold performance metrics are used to demonstrate how the newly proposed LHFGSO-DMN performed.

4.6.1 Analysis with Training Data Percentage

Figure 5 depicts a comparison of the Proposed LHFGSO-DMN using different evaluation metrics and training data percentages. A comparative investigation of the Proposed LHFGSO-DMN for accuracy is revealed in Fig. 5a With training data is 90%, the accuracy of Proposed LHFGSO-DMN is 94.63%, DCNN is 76.05%, Panoptic technique is 79.71%, Focal-Net has 87.93% and ResNet has 89.40%, and HFGSO based DRN is 92.63%. The Proposed LHFGSO-DMN attained superior accuracy, which is 5.52% better than the ResNet method. The evaluation based on sensitivity is referred to in Fig. 5b Here, analysing 80% of training data, sensitivity measured for different models such as Focal-Net, Panoptic model, DCNN, ResNet, HFGSO based DRN, Proposed LHFGSO-DMN is

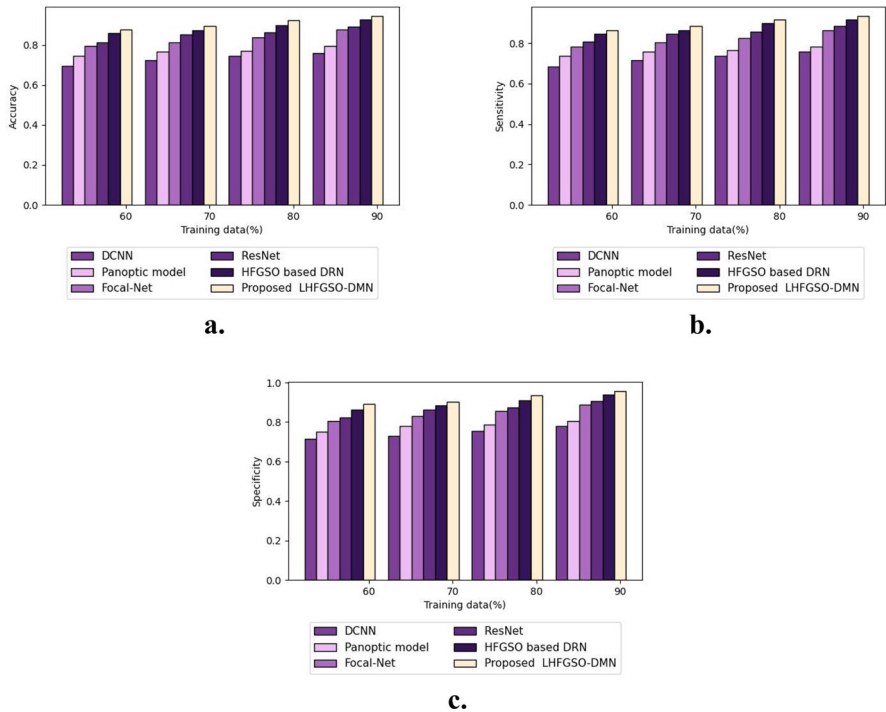


Fig. 5 Comparative analysis of proposed LHFGSO-DMN based on **a** accuracy, **b** sensitivity, **c** specificity

82.47%, 73.65%, 76.49%, 85.68%, 89.71% and 91.56%. The Proposed LHFGSO-DMN has achieved a sensitivity that is 6.42% higher than that of the Focal-Net. Specificity-based assessment of the LHFGSO-DMN is depicted in Fig. 5c here, analysing 90% of training data, the specificity of DCNN is 78.01%, the Panoptic method is 80.43%, the Focal-Net is 88.95%, ResNet is 90.76%, HFGSO-based DRN is 93.88%, and proposed LHFGSO-DMN is 95.72%. The proposed LHF-GSO-DMN derived superior specificity, which is 15.97% better than the panoptic model.

4.6.2 K-Fold Data Based Analysis

Figure 6 illustrates the relationship between K-fold data and performance metrics concerning the Proposed LHFGSO-DMN K-fold value and other performance metrics. Figure 6a depicts the comparative analysis of Proposed LHFGSO-DMN's accuracy. For k-fold is 7, the accuracy value is 76.42% of DCNN, the Panoptic model has 78.80%, the Focal-Net is 85.50%, ResNet is 87.91%, HFGSO-based DRN is 91.92% and Proposed LHFGSO-DMN is 94.06%. The Proposed LHFGSO-DMN has good accuracy, which is 2.27% higher than HFGSO-based DRN. Figure 6b indicates the comparative assessment of Proposed LHFGSO-DMN for sensitivity. The sensitivity of DCNN, HFGSO-based DRN, Panoptic technique, ResNet, Focal-Net and Proposed LHFGSO-DMN is 75.89%, 91.05%, 77.05%, 86.64%, 84.51% and 93.13% for

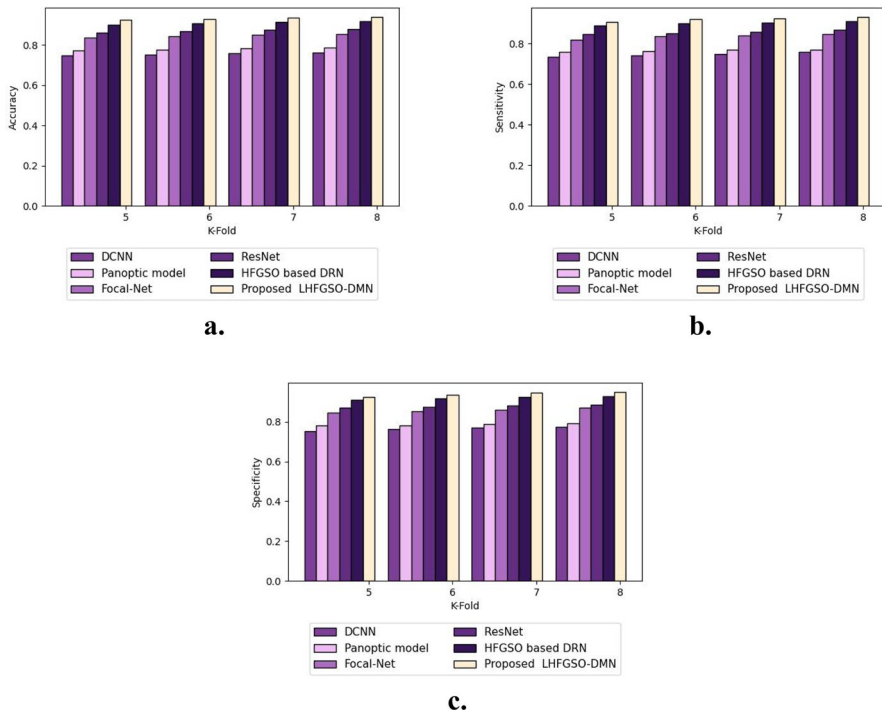


Fig. 6 Comparative analysis of proposed LHFGSO-DMN **a** accuracy, **b** sensitivity, **c** specificity

k-fold value 7. The performance improvement of the proposed LHFGSO-DMN is 6.96% better compared to the ResNet method. Figure 6c explains the comparative analysis of the proposed LHFGSO-DMN for specificity. When the K-fold value is 7, the specificity of Proposed LHFGSO-DMN is 94.99% while DCNN is 77.54%, the Panoptic model is 79.12%, Focal-Net has 87.06%, ResNet is 88.56%, and HFGSO-based DRN is 94.99%. The proposed LHFGSO-DMN is 8.34% better compared to Focal-Net.

4.7 Performance Analysis

The performance metrics of the newly introduced Proposed LHFGSO-DMN were analyzed by shifting training data.

4.7.1 Training Data Based Performance Analysis

Figure 7 indicates that shifting training data percentages makes the proposed LHFGSO-DMN's performance analysis dependent on various evaluation metrics. Figure 7a shows a precise performance analysis of the proposed LHFGSO-DMN. The accuracy of Proposed LHFGSO-DMN based on iteration 5, 10, 15 and 20 is 88.43%, 89.42%, 91.86%, and 92.78%, while analyses 90% training data value. %. Figure 7b

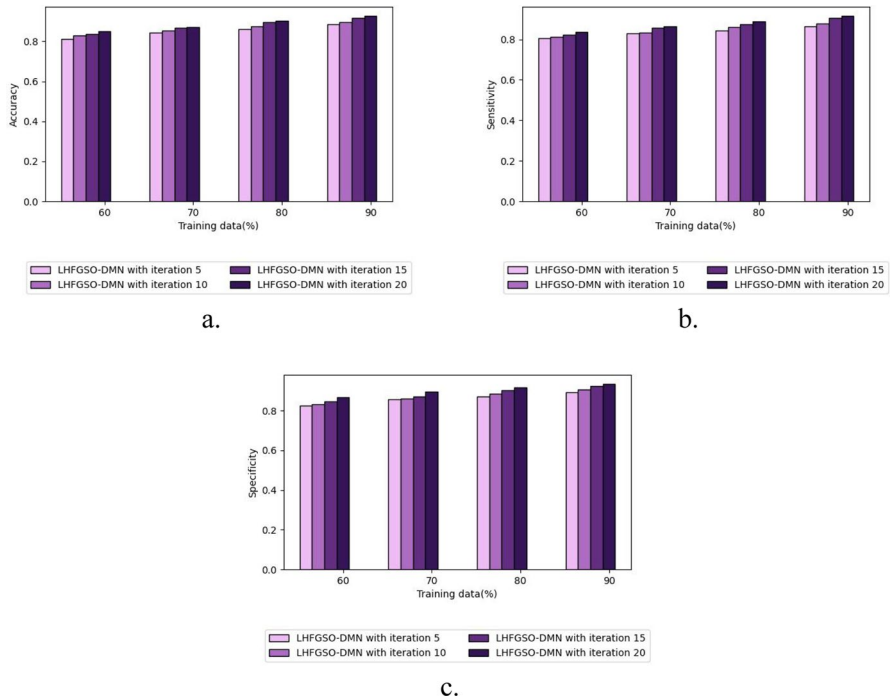


Fig. 7 Performance evaluation of proposed LHFGSO-DMN **a** accuracy, **b** sensitivity, **c** specificity

Table 2 Comparative discussion

Variations	Evaluation parameters	DCNN	Panoptic method	Focal-net	ResNet	HFGSO-based DRN	Proposed LHFGSO-DMN
Training data	Accuracy (%)	76.05	79.71	87.93	89.40	92.63	94.63
	sensitivity (%)	75.65	78.37	86.48	88.47	91.57	93.46
	Specificity (%)	78.01	80.43	88.95	90.76	93.88	95.72
K-fold	Accuracy (%)	76.42	78.80	85.50	87.91	91.92	94.06
	Sensitivity (%)	75.89	77.05	84.51	86.64	91.05	93.13
	Specificity (%)	77.54	79.12	87.06	88.56	92.86	94.99

Bold values denote higher value

specifies evaluation assessment of the proposed technique with sensitivity. By analysing training data is 90%, the sensitivity of Proposed LHFGSO-DMN with iteration 5 is 86.43%, iteration 10 is 87.96%, iteration 15 is 90.58% and iteration 20 is 91.67%. Figure 6c shows the Proposed LHFGSO-DMN specificity performance analysis. The Proposed LHFGSO-DMN has a specificity of 89.42% for 90% of the training value at Epoch 5, 90.53% at Epoch 10, 92.56% at Epoch 15, and 93.56% at Epoch 20.

4.8 Comparative Discussion

Table 2 indicates the comparative discussion of the proposed technique and the values are depicted with training data with 90% and k-fold value of 9. For analysing 90% of training data, the accuracy of the proposed LHFGSO-DMN is 94.63%, existing techniques like, DCNN is 76.05, the Panoptic model is 79.71%, Focal-Net is 87.93% ResNet is 89.40%, and HFGSO-based DRN is 92.63%. Statistical features in feature extraction result in a significant increase in the accuracy of the developed PCa detection model. The sensitivity of the Panoptic technique, DCNN, Focal-Net, HFGSO-based DRN, ResNet and Proposed LHFGSO-DMN is 78.37%, 75.65%, 86.48%, 91.57%, 88.47% and 93.46%, 90% of training data analysing. The introduced PCa detection's specificity is effectively enhanced by the multi-objective SegNet approach. By analysing 90% of training data, the specificity of DCNN has 78.01%, the Panoptic technique is 80.43%, Focal-Net has 88.95%, ResNet is 90.76% HFGSO-based DRN is 93.88%, and Proposed LHFGSO-DMN is 95.72%. By using a hybrid optimization algorithm for the Proposed LHFGSO-DMN training process, the developed PCa detection method achieved more specificity.

4.9 Confusion Matrix

The confusion matrix is the name given to the process of combining the classification model's prediction results. It displays the actual and expected label distribution for each group. It is employed in the model's performance evaluation. Here, the total number of accurate and inaccurate forecasts is divided by each class and summarised with values. The confusion matrix of the research methodologies is displayed in Fig. 8.

4.10 Convergence Curve

The convergence curve is shown in Fig. 9. When iteration is 25, the fitness value of the existing methods, such as SGD optimizer, Adam optimizer, RMSprop optimizer, HFGSO, and Proposed LHFGSO are 0.0571, 0.0458, 0.0314, 0.0205, and 0.0252, respectively.

5 Conclusion

This paper introduced, the Proposed LHFGSO-DMN for effective early detection of PCa, to reduce the death of affecters. To detect prostate cancer, the MRI image is obtained from the prostate MRI database. In addition to enhancing the worth of the input image, the input image is pre-processed using an adaptive median filter and ROI is extracted. The essential region segmentation in segmentation is then performed using the multi-objective SegNet model. In this instance, where SegNet is

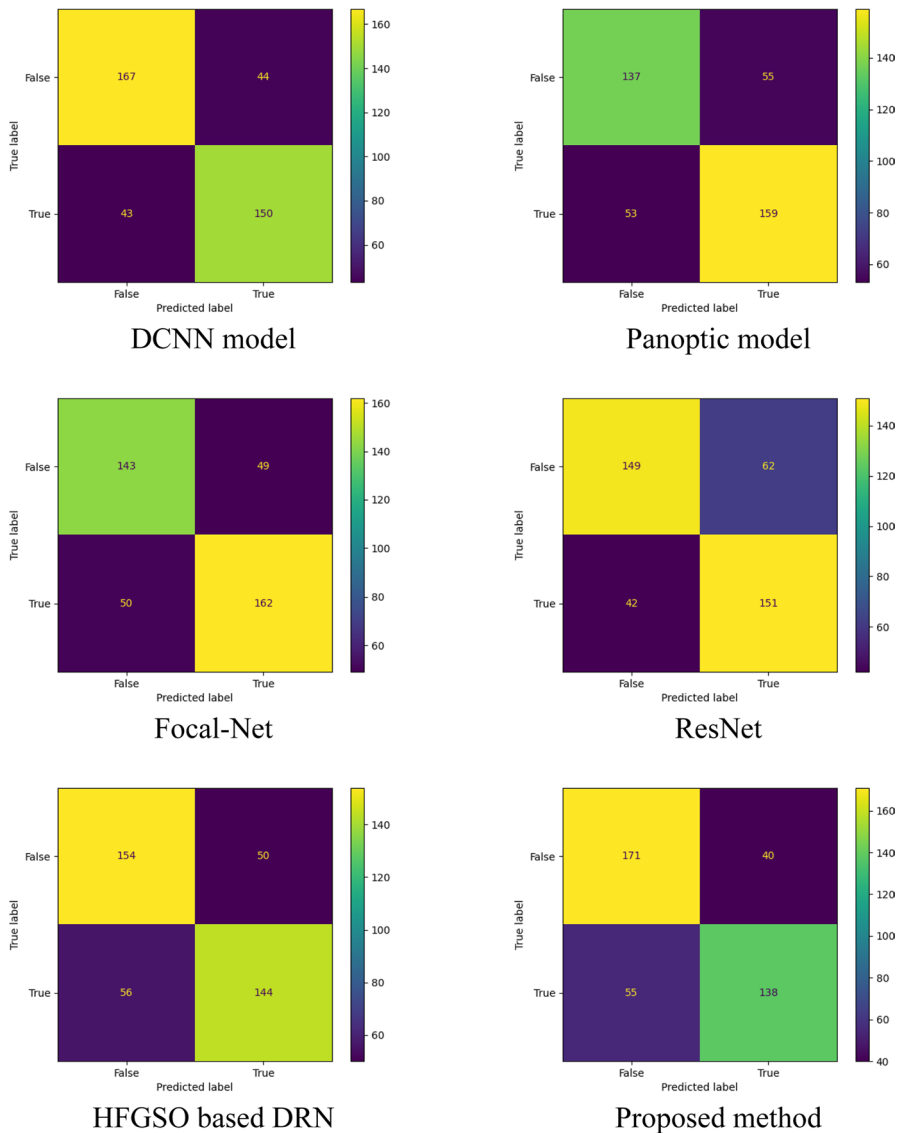


Fig. 8 Confusion matrix

trained by LHFGSO, loss functions and like, dice coefficient and pixel-wise entropy are also taken into account. Rotation, cropping, flipping, and random erasing are the methods used for data augmentation following the completion of segmentation. Then, the features are extracted based on the LBP, SLBT, and statistical features. Finally, DMN is used to train by the developed LHFGSO method for PCa detection. The proposed PCa detection recorded the highest accuracy value of 94.63%, a sensitivity of 93.46%, and a specificity of 95.72% the proposed LHFGSO-DMN achieves

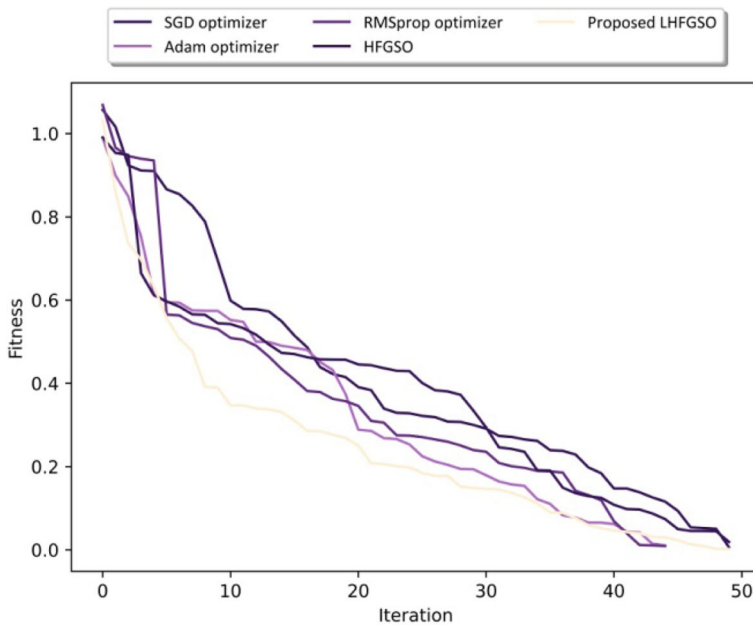


Fig. 9 Convergence curve

better performance. In future dimensions, additional features will be taken into consideration to further enhance the developed PCa detection model.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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